

Draft - expert review pending

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Topic 1.3 - Ecosystems and natural resources

A real classroom situation

Every year, teachers from Hyderabad take their pupils on a trip to Keenjhar Lake, the large freshwater lake about two hours away. The pupils see fishermen casting nets, birds wading in the shallows, and thick green plants at the edge. One pupil asks, "If the fishers catch fish every day, why are there still fish?" The teacher turns the question back to the class: "What do the fish need to keep being fish?" By the end of the discussion the pupils have listed food, clean water, shelter and young, and one of them says, "So the lake has to stay a lake."

That is an ecosystem: a living community together with its non-living surroundings, functioning as one system. This topic explains ecosystems, why they matter, what threatens them and how they are conserved, and then turns to the natural resources people take from them.

Explanation

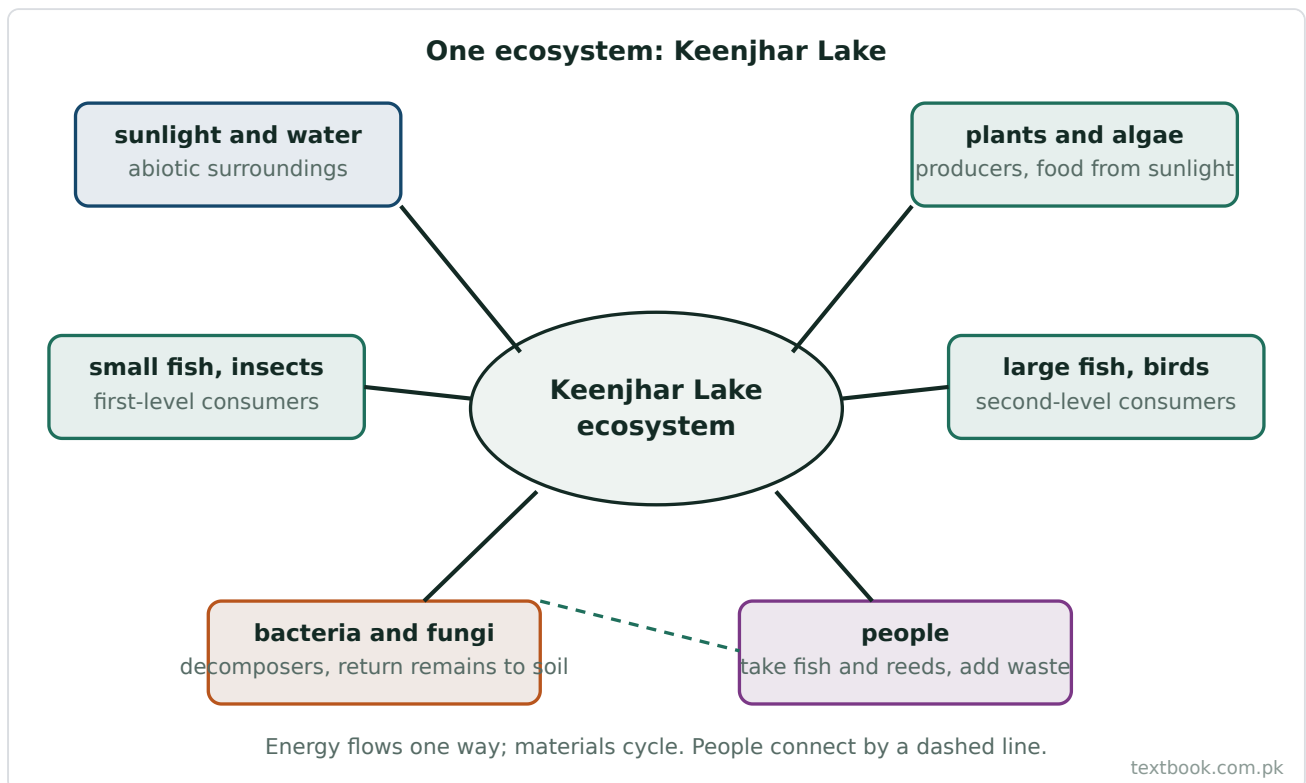
Ecosystems: types, importance, threats and conservation

An ecosystem has two kinds of parts. The living parts, plants, animals, fungi and micro-organisms, are its biotic community; the non-living parts, water, soil, air, sunlight and temperature, are its abiotic surroundings. Energy from sunlight enters through plants, which make their own food; animals eat plants or each other; decomposers return the remains to the soil. This one-way flow of energy

and the cycling of materials are what make an ecosystem a system rather than a list of species.

Ecosystems come in broad types. Forests, grasslands, deserts, freshwater lakes and rivers, coastal waters and wetlands each carry a characteristic community. Pakistan holds examples of nearly all of them: the conifer forests of the north, the thorn forests of the Indus plains, the Thar desert, the rivers and lakes of Sindh, and the mangrove creeks of the Indus Delta. A national assessment using satellite imagery ranks the Indus Delta as the world's fifth-largest mangrove forest and has mapped how Pakistan's cover changed over time (Abbas et al., 2012), which is itself an example of environmental science at work.

Why ecosystems matter has a name: ecosystem services, the benefits people receive from functioning ecosystems. The mangroves are a full worked example. Their roots hold the delta soil against the sea and soften storm waves, which protects coastal villages. Their fallen leaves feed the creek food web that supports fish and shrimp catches. Their trees shelter young fish and birds. And they store carbon. The World Health Organization lists clean air, fresh water, food security, medicines and disease and climate regulation among the services biodiversity underpins, and reports that about one million species now face extinction risk (WHO, 2025).



The threats are the human activities from Topic 1.2, arriving at once. Habitat loss, when a wetland is drained or a forest cleared; overuse, when fish, reeds or timber are taken faster than they renew; pollution, when waste enters the water or air; invasive species, when a foreign plant or animal displaces native ones; and climate change, which shifts temperature and rainfall faster than communities can move. The WHO reports that 35 per cent of the world's wetlands were lost between 1970 and the present (WHO, 2025), a warning directly relevant to Sindh's lakes.

Conservation answers each threat with a matching tool. Protected areas, such as national parks and wildlife sanctuaries, hold habitat. Sustainable use rules, such as fishing seasons and mesh-size limits, slow overuse. Pollution control, the subject of later units, removes the stress. Restoration rebuilds what was lost, which is the explicit aim of the United Nations Decade on Ecosystem Restoration, 2021 to 2030 (UN, 2021). Conservation is not the protection of nature from people; it is the management of the relationship between them.

Natural resources: types, services, issues and management

A natural resource is anything from the environment that people use to meet their needs. The basic sorting is by renewal.

- Renewable resources renew themselves on a human timescale if not overused: water in the Indus system, sunlight falling on Thar, wind, the fish in the lakes, the soil itself, and the forests and reeds that regrow.
- Non-renewable resources exist as fixed stocks that renewal cannot rebuild on any human timescale: Thar's coal, Sindh's natural gas, rock salt from the Salt Range, and groundwater in deep aquifers laid down over thousands of years.

The sorting matters because it sets the management question. For a renewable resource, the question is the rate of use: stay below the rate of renewal and the resource lasts; exceed it and it behaves like a non-renewable one on its way out. For a non-renewable resource, the question is the order and speed of use, and what replaces it when the stock ends. The Indus system is the national example: researchers writing in an open-access hydrology journal analyse its water management as a sustainability problem under changing climate and socio-

economic conditions (Archer et al., 2010), which is exactly the rate question at basin scale.

Pakistani natural resources: renewable or not?

Resource	Renewable?	The issue in one line	Management response
Indus water	yes, at a rate	withdrawals already highly stressed	stay below renewal: quotas
lake fish	yes, at a rate	catch can exceed breeding	season and mesh limits
Thar sunlight	yes	collection cost, not supply	solar panels off-grid
soil	yes, slowly	salinity and erosion strip it faster	drainage, cover crops
Thar coal	no	fixed stock; burning adds carbon	plan what replaces it
natural gas	no	wells deplete; imports fill the gap	order use: industry first

The management question: rate of use for renewables (tinted rows),
order and replacement for non-renewables (accent rows).

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A worked example. Groundwater in lower Sindh. Rainfall is thin, so the water under the fields is old water, recharged slowly. Where tube wells pump faster than recharge, the water table falls and wells go dry or salty. The resource is renewable in name and non-renewable in practice, because of the rate. Management, metering, drip irrigation, recharge ponds, crop choice, is exactly the discipline of keeping use below renewal.

Common misconception. "Resources are infinite because nature renews them." Nature renews some things, at set rates, and renews others not at all. The WHO's figure of over 2 billion people living in water-stressed countries (WHO, 2023) describes what happens when use runs ahead of renewal on the most renewable resource of all. Renewability is a rate, not a permission.

Activity: Resource rate debate

Work in two groups. One group represents a fishing community on Keenjhar Lake; the other represents a factory that wants the lake's water for cooling.

Each group lists what it needs, at what rate, and what happens to the other side if it takes that rate. Then find one rule both groups could accept, such as a catch limit, a season, or a closed zone. Allow 25 minutes, then each side presents the rule it accepted and why.

Check your understanding

1. Define an ecosystem and name its two kinds of parts. (*Remember*)
2. List three ecosystem services of the Indus Delta mangroves and explain how each protects or feeds people. (*Understand*)
3. Why is groundwater in Sindh behaving like a non-renewable resource? Name the rate that decides the answer. (*Analyze*)
4. Classify these as renewable or non-renewable: Indus water, Thar coal, lake fish, natural gas. Give one management rule for the renewable ones. (*Apply*)

Summary

An ecosystem is a living community plus its non-living surroundings, joined by a one-way flow of energy and the cycling of materials. People receive ecosystem services from functioning ecosystems: protection, food, water, climate regulation. The threats are habitat loss, overuse, pollution, invasive species and climate change, and conservation answers each with protected areas, sustainable use rules, pollution control and restoration. Natural resources sort into renewable and non-renewable, and the management question differs: rate of use for renewables, order and replacement for non-renewables. Groundwater shows how a renewable resource turns non-renewable when use exceeds renewal.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define an ecosystem and name its biotic and abiotic parts.
- ☐ I can give three ecosystem services with a Pakistani example of each.

- ☐ I can name the five main threats to ecosystems and match a conservation tool to each.
- ☐ I can classify a resource as renewable or non-renewable and state the management question that follows.

Try this at your practicum school

Make a one-page resource map of the school and its street. Mark one renewable resource the school uses (water, sunlight, a tree's shade) and one non-renewable one (gas, grid electricity from fossil plants, a concrete source). Under each, write the rate question: how much, how fast, from where. Show the map to the school's clerk or caretaker and note one thing they monitor that you had not thought of.

Summative task

Write 180 to 220 words on the following.

The school where you did your practicum sits inside a real ecosystem and uses real resources. Identify the ecosystem your school belongs to, name two services it provides, and analyse one resource the school uses: is it renewable, at what rate is it used, and what would a sustainable rule for it look like?

Mini-rubric

Criterion	What a strong response does
Ecosystem identified (Understand)	Names the actual local ecosystem, not "nature"
Services named (Understand)	Gives two services with a concrete link to the school
Resource analysed (Analyze)	Classifies the resource, states the rate question and proposes a workable rule

Criterion	What a strong response does
Clarity and register	Plain sentences; within the word count

Further reading

- Abbas, S., Qamer, F. M., Hussain, N., Saleem, R., & Nitin, K. T. (2012). National level assessment of mangrove forest cover in Pakistan. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XXXVIII-8/W20*, 187-192.
<https://doi.org/10.5194/isprsarchives-XXXVIII-8-W20-187-2011>
- Archer, D. R., Forsythe, N., Fowler, H. J., & Shah, S. M. (2010). Sustainability of water resources management in the Indus Basin under changing climatic and socio economic conditions. *Hydrology and Earth System Sciences, 14*(8), 1669-1680. <https://doi.org/10.5194/hess-14-1669-2010>
- World Health Organization. (2025). *Biodiversity* (fact sheet). Ecosystem services and the extinction-risk status: <https://www.who.int/news-room/fact-sheets/detail/biodiversity-and-health>
- United Nations. *UN Decade on Ecosystem Restoration (2021-2030)*: <https://www.decadeonrestoration.org/>
- Haines, A., & Frumkin, H. (Eds.). *Planetary Health*. Island Press. (guide-recommended book, cited by reference only)

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